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B.Tech. Degree V Semester Regular/Supplementary Examination January 2023

ME 19-205-0505 POWER PLANT ENGINEERING (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

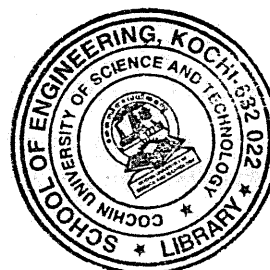
Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Get a general understanding of the scale and scope of hydroelectric power stations and also grasp the basics of diesel engine and gas turbine power plants.
- CO2: Comprehend the complexity of the steam power plant after getting to know about the working of its different components and elements.
- CO3: Understand the basic principle of the working of nuclear power plant and the various types of nuclear reactors with its safety aspects, especially with respect to the current scenario of national and international debates on the same.
- CO4: Equip with the knowledge of the basics of various non-conventional energy producing methods.
- Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
- PO – Programme Outcome

PART A (Answer *ALL* questions)

- | | | (8 × 3 = 24) | Marks | BL | CO | PO |
|--------|--|--------------|-------|----|----|---------|
| I. (a) | What are the different types of loads connected to a Power supply system? | 3 | 3 | L1 | 1 | 1,2,3,4 |
| (b) | The incremental fuel costs for two generating units a and b of a Power Plant are given by the relations:
$dF_a/dP_a = 0.085 P_a + 30$ $dF_b/dP_b = 0.095 P_b + 25,$ Where F is the fuel cost in Rupees and P is the power output in MW? Estimate the economic loading of the two units when the total load supplied by the power plant is 250 MW. | 3 | 3 | L4 | 2 | 1,2,3,4 |
| (c) | Explain two methods used for starting Diesel Engines? (b) Explain inter cooling in Gas Turbines. | 3 | 3 | L2 | 2 | 1,2,3,4 |
| (d) | Draw the P-v and T-s diagram of an ideal Brayton cycle. Show changes in T-s or p- V diagram in case of process irreversibility. | 3 | 3 | L2 | 3 | 1,2,3,4 |
| (e) | Explain the functions of any three boiler accessories. | 3 | 3 | L3 | 2 | 1,2,3 |
| (f) | What is the function of Moderators in Nuclear Power Plants? What are different materials used as Moderators and Coolants in Nuclear Power Plants? | 3 | 3 | L3 | 2 | 1,2,3 |
| (g) | What is the average value of solar insolation? Name two instruments used to measure solar insolation. | 3 | 3 | L1 | 2 | 1,2,3 |
| (h) | Explain the principle of Thermionic Energy generation. | 3 | 3 | L1 | 3 | 1,2,3 |



(P.T.O.)

PART B

(4 × 12 = 48)

- II. A central station is supplying energy to a community through two substations. Each substation feeds four feeders. The maximum daily recorded demands are:

POWER STATION..... 12,000 KW

Substation A 6000 kW	Sub-station B 9000 kW
Feeder 1 1700 kW	Feeder 1 2820 kW
Feeder 2 1800 kW	Feeder 2 1500 kW
Feeder 3 2800 kW	Feeder 3 4000 kW
Feeder 4 600 kW	Feeder 4 2900 kW

Calculate the diversity factor between

(i) substations

4 L4 4 1,2,3

(ii) feeders on substation A

4 L4 4 1,2,3

(iii) feeders on sub-station B

4 L4 4 1,2,3

OR

- III. A power station supplies the following loads to the consumers:

Time (hours)	0-6	6-10	10-12	12-16	16-20	20-22	22-24
Load (MW)	30	70	90	60	100	80	60

(i) Draw the load curve and estimate the load factor of the plant.

4 L4 4 1,2,3

(ii) What is the load factor of a standby equipment of 30 MW capacity if it takes up all loads above 70 MW? What is its use factor?

4 L4 4 1,2,3

(iii) Select suitable generator units from the 20000, 25,000 and 40000 kVA assuming a power factor of 80%.

4 L4 4 1,2,3

- IV. (a) Draw the schematic diagram of a Diesel Engine Power Plant and explain its working?

8 L3 4 1,2,3,4

(b) Explain the Principle of reheating in Gas Turbines with help of T-s Diagram.

4 L3 4 1,2,3,4

OR

- V. (a) In a Gas Turbine working on Brayton cycle air-standard Brayton cycle the air enters the compressor at 0.15 MPa and 15°C. The pressure leaving the compressor is 1.1 MPa, and the maximum temperature in the cycle is 1100°C. If a regenerator with effectiveness 0.9 is incorporated into the cycle, determine:

(i) The pressure and temperature at each point in the cycle.

4 L4 6 1,2,3,4

(ii) The compressor work, turbine work, and cycle efficiency. Assume the isentropic compressor efficiency of 80% and isentropic turbine efficiency of 85%.

4 L4 6 1,2,3,4

- (b) Give two reasons as to why Diesel Engine Power Plants are used as Peak Load Power Plants generally.

4 L1 4 1,2,3,4,5

(Continued)

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		Marks	BL	CO	PO
VI.	(a) Describe Boiling water reactor (BWR) with a neat sketch.	6	L3	3	1,2,3,4,5
	(b) Explain any three types of mechanical stokers.	6	L2	2	1,2,3,4,5
OR					
VII.	(a) Explain the fluidized bed combustion.	4	L3	3	1,2,3,4
	(b) The coal consumption of a boiler is 1700 kg/hr. The temperature of the air supplied is 330 K, and the average temperature of the flue gas leaving out of the chimney is 650 K. The 33 m high chimney produces a draught of 1.9 cms of water column. Determine:				
	(i) Quantity of air supplied per kg of coal.	2	L4	6	1,2,3,4
	(ii) Draught in terms of column of hot gas	3	L4	6	1,2,3,4
	(iii) Base diameter of the chimney, assuming that 20% of theoretical draught is used for creating the flow velocity of gases through the chimney.	3	L4	6	1,2,3,4
VIII.	(a) Explain the working of				
	(i) Flat Plate Solar Collector	3	L3	3	1,2,3,4
	(ii) Evacuated Tube Solar Collector	3	L3	3	1,2,3,4
	(b) Explain the working of Magneto Hydro Dynamic Power Generation.	6	L3	3	1,2,3,4
OR					
IX.	(a) What is the maximum percentage of power contained in wind that can be extracted by a wind turbine? Show how it is determined.	6	L3	2	1,2,3,4
	(b) Explain the process of power generation in Ocean Thermal electric conversion (OTEC).	6	L2	2	1,2,3,4

Bloom's Taxonomy Levels

L1-10.8%, L2-15%, L3-38.3%, L4-35.8%
